

Lifecycle Policies and Cost Optimization

Task 1

Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.

Steps followed:

1. Logged in to the AWS Management Console and opened the S3 service.
2. Clicked "Create bucket", named it "my-zomato-images", set the Region to Asia Pacific (Mumbai), and left the remaining settings at default.
3. Clicked "Create bucket" and opened the new bucket.
4. Clicked "Upload" and added three sample image files representing menu items: "paneer-tikka.jpg", "veg-biryani.jpg", and "chocolate-brownie.jpg".
5. Left storage class as "Standard" (the default, appropriate for frequently viewed menu images) and clicked "Upload".
6. Confirmed all three images appeared in the bucket's Objects list once the upload completed.

Task 2

Set up a lifecycle policy on your 'my-zomato-images' S3 bucket to automatically transition all objects older than 30 days to the Glacier storage class.

Steps followed:

1. Opened "my-zomato-images" → "Management" tab → "Lifecycle rules" → "Create lifecycle rule".
2. Named the rule "TransitionToGlacier-30Days" and chose "Apply to all objects in the bucket" as the scope.
3. Under "Lifecycle rule actions", checked "Move current versions of objects between storage classes".
4. Added a transition: Storage class = "Glacier Flexible Retrieval", Days after object creation = 30.
5. Reviewed the rule summary, which confirmed that any object in the bucket older than 30 days would automatically move from S3 Standard to Glacier, and saved the rule.
6. Confirmed the new rule appeared as "Enabled" in the Lifecycle rules list, with the 30-day Glacier transition visible in its details.

Task 3

Simulate a cost optimization scenario: calculate and compare the estimated monthly storage cost for keeping 5GB of images in S3 Standard versus S3 Glacier using AWS Pricing Calculator.

Steps followed:

1. Opened the AWS Pricing Calculator and created a new estimate.
2. Added an "Amazon S3" service, selected the Asia Pacific (Mumbai) region, chose "S3 Standard" storage class, and entered 5 GB as the amount of storage.
3. Noted the estimated monthly cost the calculator returned for S3 Standard.
4. Added a second S3 estimate in the same region, this time selecting the "S3 Glacier Flexible Retrieval" storage class, again for 5 GB.

5. Noted the estimated monthly cost for Glacier and compared the two figures.

Estimated results (approximate, per AWS Pricing Calculator, Mumbai region):

- S3 Standard — 5 GB: about \$0.125/month $\approx$ ₹10–11/month
- S3 Glacier Flexible Retrieval — 5 GB: about \$0.02/month $\approx$ ₹1.5–2/month
- Approximate monthly saving by moving 5GB from Standard to Glacier: roughly ₹8–9/month, i.e. Glacier costs about 80–85% less than Standard for the same amount of storage.

(Exact figures vary slightly with the USD-INR conversion rate used and any pricing updates, but the relative gap — Glacier being a small fraction of Standard's cost — stays consistent, which is why lifecycle policies that move older, rarely accessed images to Glacier meaningfully cut storage costs at scale.)

Task 4

Add a new rule to your S3 bucket's lifecycle policy so that any objects with the prefix 'offers/' (e.g., 'offers/summer2024.jpg') are deleted automatically after 60 days.

Steps followed:

1. Went back to "my-zomato-images" → "Management" → "Lifecycle rules" → "Create lifecycle rule".
2. Named the new rule "ExpireOffers-60Days".
3. Under scope, chose "Limit the scope of this rule using one or more filters" and set Prefix = "offers/", so the rule only applies to objects stored under that folder path (e.g., offers/summer2024.jpg).
4. Under "Lifecycle rule actions", checked "Expire current versions of objects".
5. Set "Days after object creation" = 60 for the expiration action.
6. Reviewed the summary confirming that only objects under the offers/ prefix would be permanently deleted 60 days after upload, while the rest of the bucket (menu images) remained unaffected, and saved the rule.
7. Uploaded a test file "offers/summer2024.jpg" and confirmed the new rule showed up correctly targeting it in the Lifecycle rules list alongside the existing 30-day Glacier transition rule.

Task 5

Use ChatGPT to generate a sample JSON configuration for an S3 lifecycle policy that transitions files to Glacier after 15 days and deletes them after 180 days. Paste the generated JSON and briefly explain each section.

AI-generated JSON lifecycle configuration:

```
{
  "Rules": [
    {
      "ID": "TransitionToGlacierAndExpire",
      "Filter": {},
      "Status": "Enabled",
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 180
      }
    }
  ]
}
```

```
}  
]  
}
```

Explanation of each section:

- "Rules" — the top-level array holding one or more lifecycle rules; a bucket can have multiple rules, each targeting different objects or prefixes.
- "ID" — a unique, human-readable name for this rule ("TransitionToGlacierAndExpire"), used to identify it in the console or API responses.
- "Filter": {} — an empty filter means this rule applies to every object in the bucket; a prefix or tag could be added here to limit the scope, as was done for the offers/ rule in Task 4.
- "Status": "Enabled" — turns the rule on; setting this to "Disabled" would keep the rule saved but inactive.
- "Transitions" — defines a storage-class change; here, after "Days": 15 from object creation, the object automatically moves to the "GLACIER" storage class.
- "Expiration": { "Days": 180 } — instructs S3 to permanently delete the object 180 days after creation, regardless of which storage class it is in by that point.

Applied this JSON to "my-zomato-images" using the "Edit lifecycle rule" → "Add rule" (with the JSON/CLI equivalent via the AWS CLI `put-bucket-lifecycle-configuration` command), so any object uploaded to this bucket now transitions to Glacier after 15 days and is deleted entirely after 180 days, in addition to the two rules created earlier.